⚔️ Trade-Offs & the TCD

Week 4 • Day 5 • Technical Architecture & Constraints

TPM Academy — TCD ships today

🎯 Day 5 Objectives

- Identify and write the **top 5 trade-offs** — Option A vs B, choice, cost, revisit trigger
- Cover at least **3 categories**: coupling vs independence, consistency vs availability, latency vs durability, speed vs robustness, generality vs simplicity
- Build the **stakeholder sign-off matrix** with real names
- Run an **integration pass** across Sections 1–6
- Cross-review with another quad and ship the Technical Concept Document (TCD) as a sibling to the Product Requirements Document (PRD)

🕑 Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Pin TCD Sections 1–4 on the wall
09:15 – 10:30	Block 1 + Activity 1	Surface the trade-offs (8–10 candidates)
10:45 – 12:00	Block 2 + Activity 2	Write the top 5 trade-offs
13:00 – 14:15	Block 3 + Activity 3	Stakeholder sign-off matrix
14:30 – 16:00	Block 4 + Activity 4 + Wrap	Integration + cross-review + sign-off

Block 1 — Surface the Trade-Offs

Mature architecture documents name the tensions explicitly

📚 Mature vs Immature Trade-Offs

❌ Immature

"We considered microservices but chose monolith."

Description, not trade-off.

✅ Mature

"Option A: modular monolith. Option B: extract reconcile into a service. We chose A. Accepted cost: future extraction needs coordinated refactor. Revisit if traffic exceeds 5× rest-of-app or a second team contributes."

The pattern: Option A vs Option B → choice → accepted cost → revisit trigger.

📚 Five Categories of Architectural Trade-Off

Category	Example tension
Coupling vs independence	Module in monolith vs new service
Consistency vs availability	Strong-consistency reads vs replica reads
Latency vs durability	Sync write vs async with retry
Speed vs robustness	Ship happy path now, harden later
Generality vs simplicity	Build the generic capability or just this case

Aim: top-5 trade-offs span **at least 3 categories**. All in one category usually missed real tensions.

👥 Activity 1 — Surface Trade-Offs

Quads • 45 minutes

Brainstorm 8–10 candidates from Sections 1–4. Categorize. Cull to 5 finalists. Rough out structure.

- 25 min: brainstorm 8–10
- 5 min: categorize
- 10 min: cull to 5
- 5 min: rough Option A / B / Choice / Cost / Trigger

⏱️ 25 + 5 + 10 + 5

Block 2 — Write the Top 5

Final TCD Section 5 prose

📝 The Trade-Off Template

```
### Trade-off N — <short name>
**Tension:** <category> — <framing in one sentence>
**Option A:** <1–2 sentences>
**Option B:** <1–2 sentences>
**Choice:** Option A.
**Why:** <2–3 sentences — defended in business + operational terms>
**Accepted cost:** <what we give up, concretely>
**Revisit trigger:** <metric or organizational change>
```

"Accepted cost: complexity" is vague. Force concreteness: *maintenance burden, debugging difficulty, on-call exposure*.

📝 Worked Trade-Off

```
### Trade-off 2 — Sync vs async write to audit
**Tension:** Latency vs durability.
**Option A:** Synchronous write to audit event store
before responding to user.
**Option B:** Async publish to Event Hubs audit stream;
consumer writes to store.
**Choice:** Option B.
**Why:** Synchronous would add 30–80ms to the reconcile
latency budget, breaking the p95 ≤ 1000ms SLO. Async
with Event Hubs durability is sufficient for SOC 2 audit
requirements. The DLQ on the consumer handles rare
write failures.
**Accepted cost:** Audit events may lag 1–10s behind user
action; a regulator querying within seconds may see
stale state.
**Revisit trigger:** A regulator or customer requires
synchronous audit visibility within < 1 second.
```

Option B leans on a dead-letter queue (DLQ) to catch rare write failures.

👥 Activity 2 — Write the Top 5

Quads • 50 minutes

- 25 min: solo drafts (each member writes 2)
- 15 min: pool + refine
- 10 min: consistency check — do trade-offs contradict each other?

⏱️ 25 + 15 + 10

Block 3 — Stakeholder Sign-Off Matrix

A constraint without an owner is a constraint nobody implements

📋 The Sign-Off Matrix

Each row pairs a constraint with an owner — down to the single sign-on (SSO) scope addition.

#	Constraint (link)	Stakeholder	Status	Next step
1	Section 1 Architecture stance	Lin (Architect)	Proposed	30-min walkthrough
2	Section 3 Threat model	Pat (Security lead)	Proposed	Send brief + 30-min walk
3	Section 3 SOC 2 retention	Sam (Compliance)	Proposed	Email + 15-min sync
4	Section 4 Latency SLO	Kim (Eng lead)	Discussed	Confirm sprint review
5	Section 4 Rate-limit policy	Jamie (Platform)	Proposed	Slack + review
6	Section 5 Trade-off 1	Lin (Architect)	Proposed	Same as #1
7	SSO scope addition	Casey (Identity)	Approved	None

🎯 Status Values (be honest)

Status	Meaning
Proposed	TPM has drafted; stakeholder hasn't reacted
Discussed	Conversation happened; no formal sign-off
Approved	Stakeholder said yes (capture how)
Blocked	Pushback; needs different proposal

Most entries should be Proposed today. That's honest. The matrix is the action list for Week 6.

👥 Activity 3 — Sign-Off Matrix

Quads • 50 minutes

- 25 min: list every constraint needing sign-off
- 10 min: assign a real person per row
- 10 min: set the "next step" for each
- 5 min: status sanity check (most should be Proposed)

⏱️ 25 + 10 + 10 + 5

Block 4 — Integration + Cross-Review + Sign-Off

Read top to bottom; pair-review; ship

✅ The Integration Checklist

Check	Pass criterion
Section 1 stance ↔ Section 2 diagrams	Stance shows as expected container layout
Section 2 integrations ↔ Section 3 threats	Every boundary covered or explicitly out-of-scope
Section 3 NFRs ↔ Section 4 SLOs	Conflicts surface as Section 5 trade-offs
Section 4 SLOs ↔ Section 5 trade-offs	Latency cost appears in Section 5 if accepted
Sections 1–5 ↔ Section 6 stakeholders	Every load-bearing decision has an owner
No fortune-cookie prose	Specific to this feature

🔎 Cross-Review Focus

1. **Is each trade-off really a trade-off**, or just a description?
2. **Does the stakeholder matrix have obvious gaps?**
3. **Are Service Level Objectives (SLOs) and trade-offs internally consistent?**

> **Reviewer rubric (for scoring):**

Dimension	Weight
Stance defensibility	20%
Threat-model rigor	20%
Component clarity	15%
SLO realism	20%
Trade-off honesty	15%
Stakeholder map	10%

👥 Activity 4 — Integration + Cross-Review + Sign-Off

Quads • 55 minutes • Wrap

- 10 min: integration pass (solo + pool)
- 30 min: cross-review with another quad (rubric)
- 15 min: adopt / defer / reject + revise + internal sign-off

⏱️ 10 + 30 + 15 · 15-min wrap-share

🎉 Week 4 Wrap

What we shipped this week

- Architecture stance with revisit trigger
- Integration map + C4 diagrams
- STRIDE threat model (six threat categories) + security non-functional requirements (NFRs)
- Three SLOs + latency budget
- Top 5 trade-offs
- Stakeholder sign-off matrix
- Integrated TCD

What opens Week 5

- **Technical Infrastructure & Modeling**
- Databases & data logic
- Cloud architecture
- REST & SOAP APIs
- Perf baselines & monitoring
- TCD becomes the spine

Bring to Monday: your TCD + PRD. The component map drives Week 5 data modeling; the SLOs drive perf baselines.

💬 Questions & Reflection

Which of your trade-offs would you change if a senior architect spent 30 minutes with you?